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Disease severity and postoperative complications after appendectomy

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ABSTRACT

Appendectomy is among the most common emergency operations performed in children and is generally well tolerated. However, postoperative complications remain an important source of morbidity, particularly in patients with complicated appendicitis. Disease severity is a major predictor of adverse postoperative outcomes. Despite this, there is no universal definition for complicated appendicitis. American College of Surgeons' National Surgical Quality Improvement Program in Pediatrics (NSQIP-P) criteria provide a proposed intraoperative framework for severity stratification that may help to improve reporting and comparison of outcomes.

The most clinically relevant postoperative complications after pediatric appendectomy include organ-space infection or intra-abdominal abscess, surgical site infection, postoperative ileus, small bowel obstruction, and rare technical complications such as stump leak. Their incidence, presentation, and management vary based on appendicitis severity, operative findings, and postoperative care pathways. Current evidence suggests that severity-based postoperative care, avoidance of unnecessary drain placement, selective imaging, early transition to oral antibiotics when appropriate, and standardized recovery protocols may reduce variation in management and improve outcomes. This review summarizes the most common postoperative complications following pediatric appendectomy, with emphasis on severity stratification, prevention, recognitions, and contemporary management strategies.

Introduction

Appendicitis is among the most common surgical condition in children, and appendectomy remains one of the most frequently performed pediatric operations. Although outcomes are generally favorable, postoperative complications remain an important source of morbidity, particularly in children with complicated appendicitis. Short-term adverse events are uncommon in uncomplicated disease, with incisional surgical site infection rates around 1%.[1] In contrast, complication rates rise substantially with increasing disease severity; one study found that overall complications increased from 10.6% to 16.4% with greater disease severity.[2,3] Intra-abdominal abscess rates in complicated appendicitis can be as high as 10-20%.[4] Other complications include superficial surgical site infection, postoperative ileus, small bowel obstruction, and less common technical complications.

Terminology used to describe appendicitis severity, however, is inconsistent across the literature. Terms such as *perforated, gangrenous, complicated*, and *uncomplicated* are frequently used, but the criteria for these labels vary across studies. This lack of standardization complicates comparisons of postoperative outcomes and limits the interpretation of published literature. Because appendicitis severity is a major determinant of postoperative complication risk, a consistent framework is vital.[5]

Disease severity is a major predictor of adverse postoperative outcomes. American College of Surgeons' National Surgical Quality Improvement Program in Pediatrics (NSQIP-P) criteria provide a more objective intraoperative framework for severity stratification and may improve reporting and comparison of outcomes across institutions. A large analysis of the NSQIP-P database proposed objective intraoperative criteria to define complicated appendicitis, including extraluminal fecalith, diffuse fibrinopurulent exudate, visible perforation, and intraperitoneal abscess (Table 1). In a cohort of 82,950 patients across 148 participating hospitals, each of these findings was independently associated with increased postoperative adverse events, with a cumulative effect observed when multiple criteria were present.[5] This framework gives a clinically relevant and reproducible method for stratifying appendicitis severity and allows more consistent comparisons across studies. Importantly, it reinforces that "complicated appendicitis" is not a singular diagnosis but exists along a spectrum of disease severity that correlates with postoperative risk. This distinction in pathology severity informs downstream decisions on antibiotic duration,

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Table 1
NSQIP pediatric intraoperative criteria for complicated appendicitis.

NSQIP Pediatric criteria	Description
Visible hole	gross perforation/visible defect in the appendix
Intra-abdominal or pelvic abscess	abscess identified at operation
Extraluminal fecalith	fecalith outside the appendiceal lumen
Diffuse fibrinopurulent exudate	fibrinopurulent contamination extending beyond the right lower quadrant/pelvis

*NSQIP = National Surgical Quality Improvement Program
*Each criterion has been independently associated with increased adverse events and resource utilization[5]
*There is a stepwise increase in risk of adverse event as additional criteria are present[5]

postoperative imaging, and intervention thresholds.

Complicated appendicitis is consistently associated with increased postoperative morbidity, including higher rates of infectious complications, readmissions, and emergency department revisits compared to uncomplicated disease.[5] However, variability in how severity is defined across datasets limits direct comparison between studies and contributes to heterogeneity in reported outcomes. Appendicitis severity is a primary indicator of postoperative complications and should be accounted for when evaluating outcomes, comparing management strategies, and developing clinical pathways. The risk, presentation, and management of postoperative complications vary substantially based on appendicitis severity, operative findings, and postoperative care pathways. This review summarizes the most common and clinically relevant complications after pediatric appendectomy, with an emphasis on severity stratification, prevention, and current management strategies.

Infectious complications

Surgical site infections (SSI) include intra-abdominal abscess (IAA) and superficial infection. Although postoperative IAA is uncommon when all appendicitis severities are considered together, its incidence increases significantly in complicated disease, where reported rates are as high as 10- 20%.[4] This difference highlights a recurring theme of the literature: disease severity, rather than appendectomy alone, is the dominant driver of postoperative abscess risk.

Reported IAA rates vary widely across studies, in part because of the differences in how complicated appendicitis is defined, the degree of intra-abdominal contamination included within study cohorts, and the threshold for postoperative imaging and diagnosis. Risk appears to increase with greater disease burden, including visible contamination, abscess at presentation, and extraluminal fecalith.[5] Longer operative duration has also been associated with higher postoperative morbidity, including IAA, although this may reflect operative complexity more than serve as an independent cause of abscess formation.[6]

Several intraoperative practices have been studied as potential modifiers of postoperative abscess risk. Routine peritoneal irrigation has not demonstrated benefit over suction alone in perforated appendicitis.[4] More recently, a randomized controlled trial-based meta-analysis trial suggested that povidone-iodine irrigation may reduce post-operative IAA formation in patients who presented with perforated appendicitis.[7] Similarly, prophylactic drain placement has not been shown to reduce postoperative abscesses and, in pooled analyses, has been associated with worse outcomes, including higher abscess rates and increased length of stay.[8] Taken together, these data do not support routine irrigation or routine drain placement as strategies to prevent postoperative IAA. Although there may be some utility in betadine irrigation based on emerging data, further investigation is required. If a drain is placed, its use should be selective and driven by specific intra-operative concerns rather than habit.

Earlier concerns that laparoscopic appendectomy increased post-operative abscess risk compared with open appendectomy have not been consistently supported by more recent analysis.[9] Contemporary meta-analyses suggest that abscess rates are similar across approaches, while laparoscopy has the advantage of a shorter length of stay and fewer wound complications.[10] As in other comparisons in the literature, disease severity remains a more important predictor of IAA than operative approach alone.[5]

Management of postoperative IAA remains heterogeneous, reflecting limited high-quality pediatric evidence. In practice, treatment is guided by clinical status, imaging findings, abscess size, and whether collections are unifocal or multifocal. Available retrospective data suggest that small, unifocal abscesses can often be treated successfully with antibiotics alone, whereas larger or multifocal collections are less reliably managed nonoperatively and more frequently require drainage.[11,12] Although drainage procedures are often necessary for larger or multifocal post-appendectomy abscesses, invasive treatments are associated with longer hospital length of stay than antibiotics alone. This reflects treatment burden and the fact that clinically more severe abscesses are preferentially selected for invasive treatment. In a recent study, drainage procedures were associated with a median LOS of 17 days compared to 7 days for non-invasive treatment.[11]

Superficial surgical site infections (SSIs) are among the more common complications following pediatric appendectomy. National data suggest superficial SSI occurs in approximately 1% to 4% of all patients, including cohorts with mixed appendicitis severity.[1,13] However, studies that stratify by disease severity have reported substantially higher rates. Non-perforated appendicitis with findings of gangrene, suppuration, or exudate has an increased risk of developing an SSI at 4.3% compared to 2.2% in patients without these findings.[13] Modifiable risk factors for superficial SSI include open compared to laparoscopic surgery, drain use, delayed operative management, and longer operative duration. Open appendectomy has a higher rate of SSI compared to laparoscopic appendectomy.[9] As with other postoperative complications, disease severity appears to be an important predictor of postoperative SSI risk.[5]

Antibiotic treatment is a central component of management for both deep and superficial surgical site infections, although the optimal route and duration remain variable across studies and institutional pathways. Data supporting an early transition from intravenous to oral antibiotics suggest that oral therapy is a safe and effective alternative in patients with complicated appendicitis, with outcomes similar to those with intravenous therapy and the benefit of a shorter length of stay.[14,15] Standardized postoperative antibiotic pathways may reduce variation in care without increasing IAA. Related pathway studies suggest that protocolized care can reduce postoperative abscess rates, although the mechanism is likely multifactorial and may extend beyond the antibiotic route alone.[16] Published pediatric protocols have compared shortened postoperative regimens of approximately 2 to 5 days of intravenous antibiotics with more traditional 5 to 10-day courses, with many pathways opting for a transition to oral therapy once patients are afebrile, tolerating a diet, and clinically improving. These protocols have shown that a shortened antibiotic regimen is noninferior and does not increase the risk of IAA compared with a longer antibiotic duration.[17]

Emerging evidence supports the use of structured risk assessment tools and severity grading scales to reduce complication rates associated with complex disease.[2] Early identification of high-risk patients may enable targeted monitoring, thereby decreasing the need for further interventions due to SSI/IAA and related complications.[16,5]

Obstructive complications

Post-operative ileus

Postoperative ileus is an uncommon complication following appendectomy. It is typically defined as delayed return of bowel function after surgery, often beyond 72 hours, and presents with abdominal distention, nausea/vomiting, and poor oral intake in the absence of a mechanical obstruction. A meta-analysis from 2006 demonstrated a lower incidence of postoperative ileus after laparoscopic compared with open appendectomy, decreasing from 2.8% to 1.3%.[18] However, contemporary studies have not consistently evaluated postoperative ileus as a discrete outcome in pediatric appendectomy populations, which limits precise estimates of its current incidence and risk factors.

Although postoperative ileus is often self-limited, it is clinically important because it contributes to prolonged hospitalization and increased resource utilization. Management is generally supportive and includes bowel rest, intravenous fluid administration, correction of electrolyte abnormalities, and monitoring for return of bowel function. In more prolonged cases, additional nutritional support may be required. Notably, many studies do not report postoperative ileus as an explicit outcome, even though return of bowel function is commonly incorporated into discharge criteria. This likely leads to under-recognition of ileus as a measurable postoperative complication.

In patients with complicated appendicitis, prolonged ileus or bowel dysfunction may contribute to the use of parenteral nutrition (PN). NSQIP-P data showed variation in parenteral utilization after appendectomy for complicated appendicitis between hospitals, with PN use ranging from 0% to 32.4% between low- and high-utilization hospitals.[19] PN therefore serves as one marker of the downstream resource burden associated with delayed recovery (not specifically with ileus alone), although its use may also reflect institutional practice patterns rather than disease severity alone.

Small bowel obstruction

Although small bowel obstruction (SBO) is an uncommon complication after appendectomy, it is among the more clinically significant postoperative adverse events because it often leads to readmissions, prolonged hospitalization, and additional imaging or interventions. Most postoperative SBOs present in a delayed fashion and are often related to adhesive disease or sequelae of complicated intra-abdominal inflammation. The overall risk of postoperative SBO is low, generally less than 1%.[20] However, complicated appendicitis is associated with a higher risk of subsequent SBO, likely reflecting greater intra-abdominal inflammation, abscess formation, and adhesive burden.[21] Major risk factors associated with post-operative SBO include complicated or perforated appendicitis and postoperative intra-abdominal abscess, both of which increase the risk of adhesive obstruction. Recent systematic review data suggest that postoperative SBO risk is higher after complicated than simple appendicitis, with reported rates of 2.9% versus 0.47%, respectively, and an odds ratio of 2.02 (1.35-2.69.[22, 23]

Earlier retrospective studies suggested that laparoscopic appendectomy might reduce the long-term risk of SBO compared with open appendectomy.[10] More recent pooled data, however, have not shown a meaningful difference in SBO risk between laparoscopic and open approaches [OR 0.96 (-.23-1.68)]. This indicates that disease severity is likely a more important predictor than operative approach along.[23] This study further demonstrated that postoperative risk was more strongly associated with complicated than simple appendicitis, reinforcing the role of intra-abdominal inflammation in subsequent adhesive obstruction.

Concern also persists that loose or retained staples may act as a nidus for postoperative bowel obstruction. However, a recent PHIS study evaluating SBO and reoperation after laparoscopic appendectomy found no meaningful association between stapler use and postoperative obstruction. Although retained staples remain a recognized anecdotal or case-based cause of early postoperative SBO, comparative pediatric data do not support stapler use itself as an independent risk factor for postoperative SBO.[24]

The operative approach is closely associated with disease severity, case complexity, and intraoperative findings, which can influence these comparisons. Patients who undergo conversion from laparoscopic to open appendectomy appear to have the highest reported incidence of later SBO, with one study reporting a rate of 1.0%.[22] Conversion to open is associated with more severe disease presentation. Reported risk factors for conversion include abscess, peritonitis, older age, obesity, and treatment at a non-pediatric hospital. A study using the Kids' Inpatient Database found an overall conversion rate of 2.2%.[25]

Distinguishing prolonged ileus from early SBO can be challenging in the immediate postoperative period, but persistent symptoms, worsening distention, or concerning imaging findings should raise suspicion for mechanical obstruction.

Rare technical complications

In the pediatric population, other complications following appendectomy have become exceedingly rare, largely due to improvements in perioperative management, risk stratification, and surgical techniques. These complications include stump related complications, bowel injury, and post-operative bleeding. Stump leak or perforation is infrequently reported and is described primarily in isolated case reports or small series, but it should remain a diagnostic consideration in patients with persistent right lower quadrant pain, severe inflammatory changes on imaging, and sepsis after appendectomy. Available pediatric studies comparing endoloop and endostapler closure techniques suggest similar overall postoperative outcomes, although true stump-related complications are too rare to determine whether one method is superior.[26] In practice, the more important factor is likely accurate identification of the appendiceal base and complete resection of appendiceal tissue, particularly in the setting of severe local inflammation or distorted anatomy. When they do occur, stump leaks can result in significant morbidity. Definitive management consists of the complete removal of the appendix with closure of the colon to prevent recurrence.

Stump appendicitis is another rare but recognized stump-related delayed complication after appendectomy and should remain a diagnostic consideration in children presenting with recurrent right lower quadrant pain after prior appendectomy. The available pediatric literature is limited largely to case reports, small series, and a recent systematic review, but suggests that stump appendicitis most often results from residual appendiceal tissue left at the index operation, particularly when severe inflammation or distorted anatomy makes identification of the true appendiceal base difficult.[27] Management is operative, with completion appendectomy or excision of the residual appendiceal remnant required for definitive treatment.

Similarly, bowel injury following appendectomy appears to be a rare complication, but its true incidence is poorly defined because it is inconsistently reported across studies. Risk is greatest in the setting of severe inflammation, perforated appendicitis, difficulty with the appendiceal base, or technical injury. Clinically, bowel injury can present with bilious vomiting, distention, obstruction and peritonitis with sepsis. Management depends on the timing of recognition, ranging from immediate primary repair to partial cecal resection or reoperation for postoperative leak or obstruction.[6]

Despite their low incidence, these complications carry significant clinical consequences, including the potential for peritonitis, need for reoperation, and prolonged hospitalization. As such, it is important to have a high index of suspicion for stump-related and technical complications intra-operatively and in the post-operative setting.

Conclusion

Appendectomy remains among the most common operations performed in children and is generally associated with good outcomes. However, postoperative complications remain clinically significant, particularly in patients with complicated appendicitis.[5] Across the literature, disease severity is the most consistent predictor of adverse postoperative events and should be considered when interpreting outcomes, counseling families, and guiding postoperative management.

A standardized intraoperative severity framework, such as the ACS NSQIP Pediatric criteria for complicated appendicitis, provides a useful structure for risk stratification and may improve data reporting and better facilitate comparisons across institutions, thereby strengthening understanding of postoperative complications. This is particularly relevant given that the most morbid postoperative complications, including IAA, ileus, and SBO, occur disproportionately in children with more complicated disease.[2]

Taken together, the available literature supports a severity-based approach to postoperative care after pediatric appendectomy. More consistent definitions, standardized care pathways, and thoughtful intraoperative severity assessment may improve postoperative counseling, reduce variation in management, and better identify patients at highest risk for complications. The greatest value of postoperative protocols is not that they prevent every complication, but that they standardize expectations for recovery and trigger earlier interventions when patients fall off the expected course. In practice, these pathways can guide antibiotic duration, diet advancement, discharge readiness, and evaluation of persistent symptoms. This promotes earlier recognition and more consistent management of complications while decreasing length of stay and resource utilization.

During the preparation of this work the author(s) used Grammarly in order to edit the grammar and writing of the article. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Serres SK, Cameron DB, Glass CC, Graham DA, Zurakowski D, Karki M, Anandalwar SP, Rangel SJ. Time to appendectomy and risk of complicated appendicitis and adverse outcomes in children. *JAMA Pediatr*. 2017 Aug 1;171(8): 740–746. https://doi.org/10.1001/jamapediatrics.2017.0885. PMID: 28628705; PMCID: PMC5710637.
2. Anandalwar SP, Cameron DB, Graham DA, Melvin P, Dunlap JL, Kashtan M, Hall M, Saito JM, Barnhart DC, Kenney BD, Rangel SJ. Association of intraoperative findings with outcomes and resource use in children with complicated appendicitis. *JAMA Surg*. 2018 Nov 1;153(11):1021–1027. https://doi.org/10.1001/jamasurg.2018.2085. PMID: 30046808; PMCID: PMC6583076.
3. Henry MC, Walker A, Silverman BL, Gollin G, Islam S, Sylvester K, Moss RL. Risk factors for the development of abdominal abscess following operation for perforated appendicitis in children: a multicenter case-control study. *Arch Surg*. 2007 Mar;142 (3):236–241. https://doi.org/10.1001/archsurg.142.3.236. discussion 241PMID: 17372047.
4. St Peter SD, Adibe OO, Iqbal CW, Fike FB, Sharp SW, Juang D, Lanning D, Murphy JP, Andrews WS, Sharp RJ, Snyder CL, Holcomb GW, Ostlie DJ. Irrigation versus suction alone during laparoscopic appendectomy for perforated appendicitis: a prospective randomized trial. *Ann Surg*. 2012 Oct;256(4):581–585. https://doi.org/10.1097/SLA.0b013e31826a91e5. PMID: 22964730.
5. McKie KA, Graham DA, Cramm SL, Saito JM, Rangel SJ. Population-based intraoperative disease severity criteria for pediatric appendicitis. *JAMA Surg*. 2024 Sep 1;159(9):1041–1050. https://doi.org/10.1001/jamasurg.2024.1759. PMID: 38959019; PMCID: PMC11223053.
6. Iglesias NJ, Arrowood R, Montgomery L, Leeper E, Tsao KJ, Iglesias JL. Operative time is independently associated with morbidity in pediatric complicated appendicitis. *J Surg Res*. 2022 Aug;276:143–150. https://doi.org/10.1016/j.jss.2022.02.045. Epub 2022 Mar 25. PMID: 35339782.
7. Egu C, Muscogliati R, Nocun W, Akil H, W U Saleh, Obidigbo B, R Aida Hakim, N A Quraishi, Salem K, Najjar E. RCT-based meta-analysis of povidone-iodine irrigation in spinal surgery: clarifying efficacy and safety. *Eur Spine J*. 2025 Oct;34(10): 4670–4679. https://doi.org/10.1007/s00586-025-09243-8. Epub 2025 Aug 8. PMID: 40779037.
8. Neville JJ, Aldeiri B. Drain placement in paediatric complicated appendicitis: a systematic review and meta-analysis. *Pediatr Surg Int*. 2023 Apr 8;39(1):171. https://doi.org/10.1007/s00383-023-05457-3. PMID: 37031267.
9. Low ZX, Bonney GK, So JBY, Loh DL, Ng JJ. Laparoscopic versus open appendectomy in pediatric patients with complicated appendicitis: a meta-analysis. *Surg Endosc*. 2019 Dec;33(12):4066–4077. https://doi.org/10.1007/s00464-019-06709-x. Epub 2019 Feb 25. PMID: 30805783.
10. Nataraja RM, Loukogeorgakis SP, Sherwood WJ, Clarke SA, Haddad MJ. The incidence of intraabdominal abscess formation following laparoscopic appendicectomy in children: a systematic review and meta-analysis. *J Laparoendosc Adv Surg Tech A*. 2013 Sep;23(9):795–802. https://doi.org/10.1089/lap.2012.0522. PMID: 24001159.
11. van Amstel P, The SML, Mulder I M, Bakx R, Derikx JPM, van Schuppen J, de Vries R, van der Kuip M, Zijp GW, Allema JH, Bijlsma TS, van Heurn LWE, Gorter RR. The management of post-appendectomy abscess in children; a historical cohort study and update of the literature. *Front Pediatr*. 2022 Jun 20;10, 908485. https://doi.org/10.3389/fped.2022.908485. PMID: 35799699; PMCID: PMC9254404.
12. Svetanoff WJ, Talukdar N, Dekonenko C, Dorman RM, Osuchukwu O, Fraser JD, Oyetunji TA, St Peter SD. Intra-abdominal abscess after appendectomy-are drains necessary in all patients? *J Surg Res*. 2020 Oct;254:384–389. https://doi.org/10.1016/j.jss.2020.05.016. Epub 2020 Jun 12. PMID: 32535257.
13. Cramm SL, Lipskar AM, Graham DA, Kunisaki SM, Griggs CL, Allukian M, Russell RT, Chandler NM, Santore MT, Aronowitz DI, Blakely ML, Campbell B, Collins DT, Commander SJ, Cowles RA, DeFazio JR, Echols JC, Esparaz JR, Feng C, Guyer RA, Hanna DN, He K, Kahan AM, Keane OA, Lamoshi A, Lopez CM, McLean SE, Pace E, Regan MD, Scholz S, Tracy ET, Williams SA, Zhang L, Rangel SJ, Eastern Pediatric Surgery Network. Association of gangrenous, suppurative, and exudative findings with outcomes and resource utilization in children with nonperforated appendicitis. *JAMA Surg*. 2022 Aug 1;157(8):685–692. https://doi.org/10.1001/jamasurg.2022.1928. PMID: 35648410; PMCID: PMC9161124.
14. Fraser JD, Aguayo P, Leys CM, Keckler SJ, Newland JG, Sharp SW, Murphy JP, Snyder CL, Sharp RJ, Andrews WS, Holcomb 3rd GW, Ostlie DJ, St Peter SD. A complete course of intravenous antibiotics vs a combination of intravenous and oral antibiotics for perforated appendicitis in children: a prospective, randomized trial. *J Pediatr Surg*. 2010 Jun;45(6):1198–1202. https://doi.org/10.1016/j.jpedsurg.2010.02.090. PMID: 20620320.
15. Wang C, Li Y, Ji Y. Intravenous versus intravenous/oral antibiotics for perforated appendicitis in pediatric patients: a systematic review and meta-analysis. *BMC Pediatr*. 2019 Nov 4;19(1):407. https://doi.org/10.1186/s12887-019-1799-6. PMID: 31684906; PMCID: PMC6827245.
16. Yousef Y, Youssef F, Homsy M, Dinh T, Pandya K, Stagg H, Baird R, Laberge JM, Poenaru D, Puligandla P, Shaw K, Emil S. Standardization of care for pediatric perforated appendicitis improves outcomes. *J Pediatr Surg*. 2017 Dec;52(12): 1916–1920. https://doi.org/10.1016/j.jpedsurg.2017.08.054. Epub 2017 Sep 5. PMID: 28935397.
17. Löfgren P, Berge E, de Wijkerslooth EML, Jivegård L, Sjöström S, Wallerstedt SM. Effectiveness and safety of shortened postoperative antibiotic regimens in children with perforated appendicitis: a systematic review and meta-analysis. *Eur J Pediatr Surg*. 2026 Jun;36(3):210–216. https://doi.org/10.1055/a-2761-5649. Epub 2025 Dec 24. PMID: 41443239.
18. Aziz O, Athanasiou T, Tekkis PP, Purkayastha S, Haddow J, Malinovski V, Paraskeva P, Darzi A. Laparoscopic versus open appendectomy in children: a meta-analysis. *Ann Surg*. 2006 Jan;243(1):17–27. https://doi.org/10.1097/01.sla.0000193602.74417.14. PMID: 16371732; PMCID: PMC1449958.
19. Kashtan MA, Graham DA, Anandalwar SP, Hills-Dunlap JL, Rangel SJ. Variability, outcomes and cost associated with the use of parenteral nutrition in children with complicated appendicitis: A hospital-level propensity matched analysis. *J Pediatr Surg*. 2021 Dec;56(12):2299–2304. https://doi.org/10.1016/j.jpedsurg.2021.03.005. Epub 2021 Mar 11. PMID: 33814183.
20. Tsao KJ, St Peter SD, Valusek PA, Keckler SJ, Sharp S, Holcomb 3rd GW, Snyder CL, Ostlie DJ. Adhesive small bowel obstruction after appendectomy in children: comparison between the laparoscopic and open approach. *J Pediatr Surg*. 2007 Jun; 42(6):939–942. https://doi.org/10.1016/j.jpedsurg.2007.01.025. discussion 942PMID: 17560198.
21. Håkanson CA, Fredriksson F, Lilja HE. Adhesive small bowel obstruction after appendectomy in children - Laparoscopic versus open approach. *J Pediatr Surg*. 2020 Nov;55(11):2419–2424. https://doi.org/10.1016/j.jpedsurg.2020.02.024. Epub 2020 Mar 2. PMID: 32192735.
22. Omling E, Salö M, Saluja S, Bergbrant S, Olsson L, Björk J, Hagander L. A nationwide cohort study of outcome after pediatric appendicitis. *Eur J Pediatr Surg*. 2021 Apr;31 (2):191–198. https://doi.org/10.1055/s-0040-1712508. Epub 2020 Jun 26. PMID: 32590867; PMCID: PMC10499502.
23. Doshi N, Bandyopadhyay S, Green M, Richardson E, Komber A, Chen SE, Shah R, Lakhoo K. The risk of adhesive bowel obstruction in children with appendicitis: a systematic review. *J Pediatr Surg*. 2024;59(8):1477–1485. https://doi.org/10.1016/j.jpedsurg.2024.03.021.
24. Zeineddin S, Hu A, Linton S, Zhang V, De Boer C, Raval MV, Abdullah F, Goldstein SD. Association between Appendiceal Stump closure method and post-operative bowel obstruction after laparoscopic appendectomy. *J Pediatr Surg*. 2023; 58(4):643–647. https://doi.org/10.1016/j.jpedsurg.2022.12.015.
25. Johnson KN, Linnaus M, Notrica DM. Conversion from laparoscopic to open appendectomy: decreased risk at dedicated children's hospitals. *Pediatr Surg Int*. 2018 Aug;34(8):873–877. https://doi.org/10.1007/s00383-018-4297-2. Epub 2018 Jun 20. PMID: 29926162.
26. Safavi A, Langer M, Skarsgard ED. Endoloop versus endostapler closure of the appendiceal stump in pediatric laparoscopic appendectomy. *Can J Surg*. 2012 Feb;55 (1):37–40. https://doi.org/10.1503/cjs.023810. PMID: 22269300; PMCID: PMC3270083.
27. Bašković M, Zaninović L, Čizmić A, Žganjer M. Stump appendicitis in children: a systematic review. *Pediatr Surg Int*. 2023 May 15;39(1):199. https://doi.org/10.1007/s00383-023-05475-1. PMID: 37188869.